

CMPE2150 Project 02

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Warning

This is a complex circuit with many potential pitfalls, some of which could be dangerous. Do not attempt to build this circuit all at once! As indicated in the discussion below, you should understand what each section of the circuit is intended to do so you can test and troubleshoot your work as you build it stage by stage. It is important in this project that you build the circuit in the order specified. After each sub-circuit is built and tested, make sure you get screen captures or photographs of the oscilloscope traces you need to demonstrate your circuit. *You must take the oscilloscope measurements for each phase of circuit development below, or you will have to rebuild the circuit.*

There may be times in your career when you will be asked to build and test a circuit designed by someone else in the company. The ability to assemble a complicated circuit, understand what it is doing, and make it work could prove invaluable to you and to your company.

The following circuit incorporates a great deal of what you have learned in your Basic Electricity and Semiconductors Courses, and takes your circuit-building and testing skills to a whole new level. As you carry out this exercise, note how many of the concepts you've learned come together to do what, from the outside, seems to be a fairly simple task—dimming a bulb over its full range of power without flickering. Also take note, mental or otherwise, of the testing and troubleshooting skills you will use as you proceed through this exercise.

There are times when you'll be testing a circuit for which the operation is well-known, and you will be provided with expected values. In this

project, you will be provided with expected waveforms (in gray-scale) which you will compare your circuit's behaviour to, and submit full-colour shots of what you see for grading.

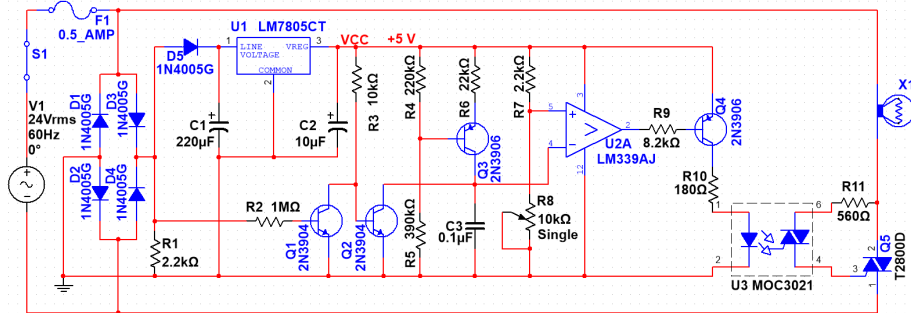


Figure 1: TRIAC Full-Power Fader—Complete Schematic

! Important

Some time after the design above was first drawn, we discovered an error. Depending on the vendor and detailed specifications of your voltage regulator, the circuit above might destroy it.

The reason is because, if you consult the [datasheet for the LM7805](#) its V_{max} rating for the DC input is 35V. This would seem to be fine for our $24V_{AC}$ source. However, recall that AC sources are generally specified as RMS voltages. So the peak voltage of the source is

$$V_P = V_{AC} \cdot \sqrt{2} = 24\sqrt{2} = 33.9V$$

You will notice that this is getting pretty close to the maximum voltage of the 7805 Voltage Regulator. Unfortunately, it gets worse. The $24V_{AC}$ transformer will be designed to give $24V_{RMS}$ even if the wall voltage is at a minimum. At NAIT's robust wall voltage, the transformer actually provides $28.7V_{AC}$, which means a peak output of

$$V_P = 28.7\sqrt{2}V = 40.6V_P$$

which—even after the voltage is dropped by three diodes ($= 2.1V_{DC}$)—is well over what our LM7805 is certified to handle. It actually worked fine for a number of years, probably because the vendor supplying our parts was providing ones robust enough to handle the excursions. However, some vendors' products have started failing over recent years. There are a number of ways of getting around this problem:

- Change from a $24V_{RMS}$ source to a lower value, although this might require a redesign of the lamp circuit.

- Add some additional diodes in series with D_5 to drop enough voltage to reduce V_p below $35V_p$. Three standard silicon signal diodes would drop the peak voltage by another $3 \times 0.7V = 2.1V$.
- Add one or two zener diodes in series with D_5 to reduce the peak voltage—Zeners typically have a reverse-bias voltage drop of several volts.
- Redesign the circuit to use a different voltage regulator.

Because you have an LM317TG Adjustable Voltage Regulator—whose [datasheet](#) specifies a V_{max} of $40V$ —we will use that.

In one of the sections below, you will **redesign the schematic above to use the LM317 Adjustable Voltage Regulator** to provide the regulated $5V_{DC}$ supply.

This is an entirely stand-alone circuit, other than the $24V_{AC}$ transformer from your kit. It contains the following sections, which you will build and test in steps:

- Full-wave bridge rectifier
- $+5V_{DC}$ regulated supply
- $120Hz$ zero-crossing detector
- Constant-current source ramp generator
- Variable analog pulse width modulator (PWM)
- TRIAC-Output optocoupler
- TRIAC power controller for a halogen lamp

Full-Wave Bridge Rectifier

In this circuit, k tely, we can have both by inserting a diode between the rectifier and the capacitor (D_5). The capacitor only draws current from the rectifier when the waveform is at a higher potential than the capacitor voltage, so as the signal amplitude falls, the diode will be reverse biased, effectively disconnecting the rectifier from the filter capacitor. R_1 acts as a load resistor for the unfiltered signal, ensuring that it pulls down to ground at the zero crossing points.

Install the following components from the schematic above:

- V_1 - $244V_{AC}$ transformer from CNT Year 2 Kit
- S_1 - SPST Toggle switch from CNT Year 1 Kit
- F_1 - $500mA$ fuse from CNT Year 2 Kit
- D_1 through D_5 - 1N400x rectifier diodes
- R_1 - $2.2k\Omega$ resistor

- C_1 - $220\mu F$ electrolytic capacitor

Notes

1. Assign two of the breadboard power rails to the switched AC signal – the output from the fuse on the top rail and the low side of the secondary to the bottom rail
2. Assign two other breadboard power rails to $+5V_{DC}$ and ground
3. Try to build your circuit as compactly as you can—It can be done on a single breadboard with some careful planning!
4. Be very careful with the orientation and connection of the diodes in the bridge
5. Be very careful with the orientation and connection of the filter capacitor—the schematic highlights the ‘+’ pin, but the ‘-’ pin is marked on the body of the capacitor; ensure that the negative pin is connected to ground
6. Do NOT connect anything on the transformer side of the bridge to ground
7. Do NOT put your oscilloscope probe ground clip anywhere on the transformer side of the bridge—put it on the ground bus

Testing

1. With the switch turned on, you should observe a full-wave signal across R_1 . Display this with oscilloscope Channel 1.
2. Across the filter capacitor C_1 , you should observe an almost completely flat DC signal when the regulator is not connected. Display this with oscilloscope Channel 2. This is what a filtered rectifier looks like with no load to draw current from the filter capacitor—the ripple voltage is effectively zero, since in the formula $V_r = \frac{I_L T}{C}$, I_L is zero.

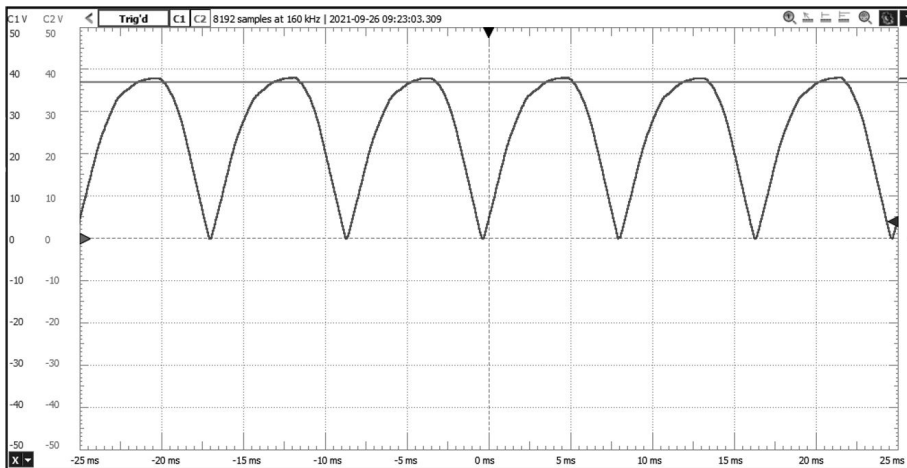


Figure 2: Rectified and Filtered Waveforms

Verify that the shape, amplitudes, and timing are satisfactory, and ask your instructor for a grade out of one mark; record the grade below. If your instructor isn't available, take a picture of the oscilloscope screen and attach it here. _____

+5V Regulated Supply

This is the section where you will modify the existing design. Replace the Voltage Rectifier subcircuit (U_1 and C_2) with an LM317 subcircuit for delivering $5V_{DC}$. Recall that, because the LM317 is adjustable, you will need to add some external resistors to set the voltage. The output decoupling capacitor C_2 should be retained. Draw your subcircuit schematic and include it here.

Testing

You should observe a constant voltage at the output pin of U_1 . Record its value here for one mark: _____

120Hz Zero-Crossing Detector

In order to synchronize the Pulse Width Modulator so that its falling edge occurs as closely as possible to 0° and 180° , we need to detect when the unfiltered full-wave rectifier signal drops to practically $0V$. Since $0.7V$ is pretty close, we will use the Base-Emitter junction of an NPN transistor to detect these events. Whenever the signal driving the Base of Q_1 through R_2 is greater than $0.7V$, Q_1 will saturate, pulling its Collector practically to zero. When the signal drops below $0.7V$, the transistor will be in cutoff, and R_3 will pull the signal HIGH. Thus, the zero crossing points will be seen as *HIGH* voltage spikes in an otherwise *LOW* signal.

Install Q_1 , R_2 , and R_3 .

Notes

1. Make sure you have the correct pinout for Q_1 .
2. In an IEEE-standard schematic, a line crossing another line is not connected unless there is a junction marker (dot). Don't make connections you shouldn't make!

Testing

With no connection to Q_2 , display the signal across R_1 using Channel 1 for reference, and the signal at the Collector of Q_1 using Channel 2. You should see something like the following. (Note the different channel settings).

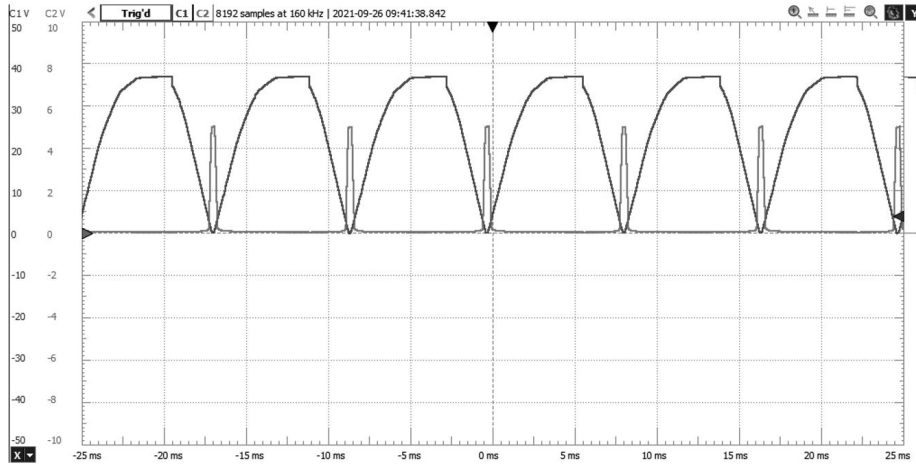


Figure 3: Zero Crossing Detector Waveforms

Once you have verified the amplitude and timing of the zero-crossing detector signal, ask your instructor to give you a grade out of one mark; record your grade here. If your instructor isn't available, attach a picture of your oscilloscope screen here. _____

Constant Current Source Ramp Generator

You've been told numerous times that transistors are practically ideal current sources. This part of the circuit uses a transistor as an ideal current source to charge a capacitor. You've also been told that, when charged by a constant current, a capacitor's voltage will increase linearly over time according to the formula $V_C = \frac{It_c}{C}$.

In our circuit, Q_2 will short capacitor C_3 to ground each time the incoming AC voltage nears zero. Theoretically, from the time between zero-crossings of the input signal, what is the charging time t_c in the formula above? _____ ms.

Q_3 , the transistor acting as a constant current source, is in a *Voltage Divider Bias* configuration, following the design requirements for a "good" circuit that were provided in your Semiconductors course. As such, its current should be reliably predictable regardless of the characteristics of the transistor. To determine the current, start by determining the Base voltage (V_b) from the voltage divider. The Emitter is at a potential

$V_{be} = 0.7V$ higher than the Base. From that, you should be able to determine the current through R_6 : _____ μA .

From this, you should be able to determine the maximum voltage expected across C_3 between zero-crossing events (it gets discharged to zero at each of these): _____ V.

Now, install the following:

- Q_2 and Q_3
- R_4 through R_6
- C_3

Notes

1. Q_3 is a 2N3906 PNP transistor. Make sure you know which pins are Emitter and Collector, both on the device and in the schematic.
2. In the schematic, watch out for a line crossing another line with no junction marker (dot). Don't make connections you shouldn't make!

Testing

With Channel 1 still on the rectifier output for reference, use Channel 2 to display the signal across C_3 . You should see something like the following:

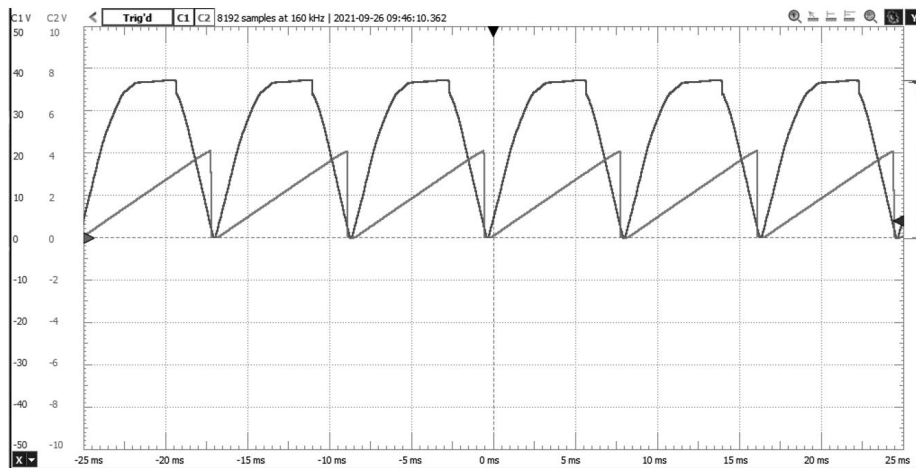


Figure 4: Constant Current Ramp Waveforms

Verify that the maximum ramp voltage matches your prediction, and that the timing is correct.

Ask your instructor to grade your work out of one mark, and record the grade assigned below. If your instructor isn't available, attach a picture of your oscilloscope screen here. _____